Wednesday 09/02/2026: Proofs

CSCI 246: Discrete Structures

Textbook reference: Sec. 5, Scheinerman

Announcements

- **Attendance Sheet** - Please sign the attendance sheet.
- **Recruiting help during group exercises** - Please grab me or Madhav (the TA) if you'd like help. Also feel free to reach out to another group. Finally, feel free to consult the textbook.
- **Weekly quiz this Friday** - It will cover definitions (Sec 3) and theorems (Sec 4). Be sure that you've done the readings and know how to do the group exercises.

Today's Agenda

- Theorems: Review group exercises on theorems ($\approx$ 15 mins)
- Proofs: Practice quiz, mini-lecture ($\approx$ 15 mins)
- Proofs: Group exercises ($\approx$ 20 mins)

Study Guide For Quiz 1

Readings (1 question)

Sec. 3 (Definitions)

- Be familiar with the definitions (understand what they say and how to apply them).

Sec. 4 (Theorems)

- Understand these truth tables: `if-then`, `iff`, `and`, `or`, `not`.
- Understand vacuous truths.

Group exercises (2 questions)

Know how to do all of the group exercises from these sections, except for the Bonus problem from Sec 4.

Follow-up

Practice Quiz

Two Proof Templates

iPAD: Review group exercises on theorems

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Read as $B \implies A$ only. Stomachaches have other causes, so *not* eating it guarantees nothing.

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"You'll get dessert if you eat that."

Read as $A \iff B$. Every child hears the half that was never said: *and if you don't, you don't.*

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Read as $A \iff B$. Every child hears the half that was never said: *and if you don't, you don't.*

English leaves the direction to **context**. Mathematics does not — which is why theorems say "if and only if" **out loud**.

Hearing "if" as "iff" is called *conditional perfection*: Geis & Zwicky, *Linguistic Inquiry* 2(4), 1971.

Resolving a puzzle from earlier

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Reminder of Definition 3.2 (Divisible)

Let a and b be integers. We say that a is *divisible* by b provided there is an integer c such that $bc = a$. The notation for this is $b|a$.

Observations

- Although $\frac{b}{a}$ is a **number**, the above shows that $b|a$ is a **proposition**:
- We don't have $0|1$ (`False`), but we do have $0|0$ (`True`).

Resolution to Puzzle

- $\frac{x}{0}$ is problematic, since not all x are divisible by 0. ($0|x$ is `False` for all integers x except $x = 0$.)
- If we take $x = 0$, we do have $0|0$, but we don't know what number $\frac{0}{0}$ equals (i.e. c in the definition could be any integer).

Follow-up

Practice Quiz

Two Proof Templates

Practice Quiz

Practice Quiz Question

Prove that the the sum of two even integers is even.
Use the appropriate proof template from the textbook.

Scheinerman Definition 3.1 (Even)

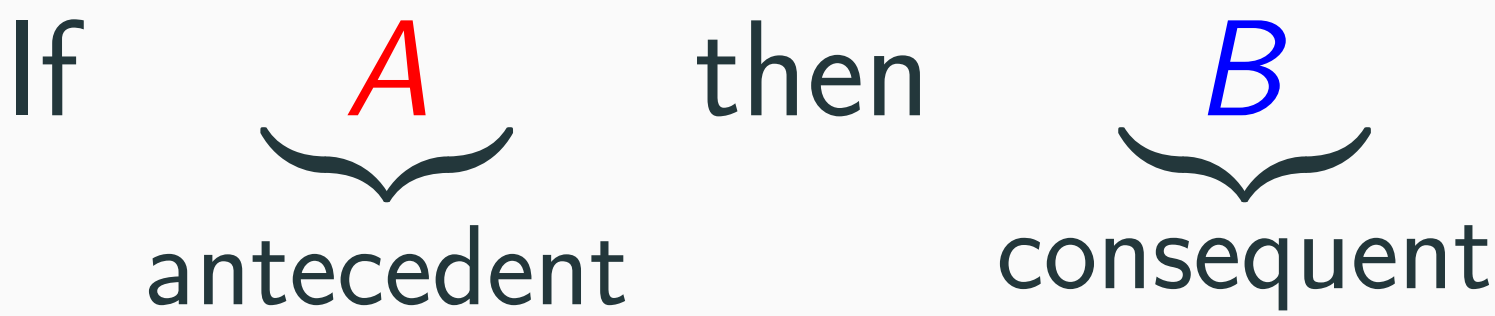
An integer is called *even* provided it is divisible by two.

Scheinerman Definition 3.2 (Divisible)

Let a and b be integers. We say that a is *divisible* by b provided there is an integer c such that $bc = a$. We also say that b *divides* a, or b is a *factor* of a, or b is a *divisor* of a. The notation for this is $b|a$.

iPad: Sketch solution

Terminology



Solution Sketch

Proposition. The sum of two even integers is even.

Proof.

Annotation	Main Text
Convert Prop. to "if-then" form	We show that if x and y are even integers, then $x + y$ is even.
State antecedent	Let x and y be even integers
Unravel defs.	Then by Defs. 3.1 and 3.2, there exist integers a, b such that $x = 2a$ and $y = 2b$.
*** The glue ***	What goes here?!?!
Unravel defs.	So there is an integer c such that $x + y = 2c$.
State consequent	Hence, $x + y$ is even.

Solution

Proposition. The sum of two even integers is even.

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State antecedent	Let x and y be even integers
Unravel defs.	Then by Defs. 3.1 and 3.2, there exist integers a, b such that $x = 2a$ and $y = 2b$.
*** The glue ***	Hence, $x + y = 2a + 2b = 2(a + b)$.
Unravel defs.	So there is an integer $c = a + b$ such that $x + y = 2c$.
State consequent	Hence, $x + y$ is even.

Follow-up

Practice Quiz

Two Proof Templates

Proof Template 1: Direct proof of an if-then theorem

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4. Unravel the definitions, working forward from the beginning of the proof and backward from the end of the proof.

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3. At the end of the proof, write down the *consequent* (the B in `if A then B`) as your conclusion.
4. Unravel the definitions, working forward from the beginning of the proof and backward from the end of the proof.
5. Forge a link between the two halves of the argument.

Proof Template 2: Direct proof of an if-and-only-if theorem

To prove a statement of the form "A iff B":

- ($\Longrightarrow$) Prove "If A, then B".
- ($\Longleftarrow$) Prove "If B, then A".

Group exercises

Random group assignments

Ashley, Isabelle: 12
Bailey, Tanna: 2
Blaisdell, Olyver: 1
Brown, Ashton: 4
Brown, Jonathan: 5
Cardenas, Dominic: 1
Cutler, George: 6
Fabik, Roland: 10
Fellows, Audrey: 7
Fitzgerald, Ryan: 11
Fuller, Ryan: 11
Griffen, Clara: 3
Grutsch, Tanner: 7
Hager, Nathan: 9
Horner, Jayden: 8
Johnson, Jaret: 6
Kintzel, Samantha: 12
Lameres, Alexis: 4
Lee, Holden: 3
Mayala, Reece: 7
McLean, Connor: 9
Mcbride, Daniel: 10
Miller, Maverick: 8
Moss, Emily: 12
Mousseau, Matthew: 4
Neuman, Evan: 3
Putnam, John: 2
Ramos, Vydel: 13
Rathbun, Jedidiah: 13
Roberts, Matthew: 8
Roller, Nathan: 11
Ryan, Otis: 1
Shevtsov, Timothy: 13
Sinclair, Aidan: 5
Smith, Charles: 2
Stensland, Emma: 10
Thiessen, Logan: 9
Thompson, Jaiden: 6
Vath, Ethan: 5

Group exercises

1. Prove that the square of an odd integer is odd.
2. Prove that the difference between consecutive perfect squares is odd.
3. Let x be an integer. Prove that $0|x$ if and only if $x = 0$.
4. Prove that an integer is odd if and only if it is the sum of two consecutive integers.

Group exercise #1: Solution

Proposition. The square of an odd integer is odd.

Proof.

Annotation	Main Text
Convert Prop. to "if-then" form	We show that if x is an odd integer, then x^2 is odd.
State antecedent	Let x be an odd integer.
Unravel defs.	Then by definition of *odd*, there is an integer a such that $x = 2a + 1$.
*** The glue ***	So $x^2 = (2a + 1)(2a + 1) = 4a^2 + 4a + 1 = 2(2a^2 + 2a) + 1$.
Unravel defs.	So there is an integer b (where $b = 2a^2 + 2a$) such that $x^2 = 2b + 1$.
State consequent	Hence, x^2 is odd.

Remark. You do not need to provide the annotations or colors in your own proofs. I am using them here in the solution to highlight the formulaic structure of an if-then proof.

Group exercise #2: Solution

Proposition. The difference between consecutive perfect squares is odd.

Proof.

Annotation	Main Text
Convert Prop. to "if-then" form	We show that if x and y are consecutive perfect squares, then $x - y$ is odd.
State antecedent	Let x and y be consecutive perfect squares
Unravel defs.	Then $x = (z+1)^2$ and $y = z^2$ where z is an integer.
*** The glue ***	So $x-y = (z+1)^2 - z^2 = (z^2+2z+1) - z^2 = 2z+1$.
Unravel defs.	So there is an integer b (where $b = z$) such that $x - y = 2b + 1$.
State consequent	Hence, $x - y$ is odd.

Remark. You do not need to provide the annotations or colors in your own proofs. I am using them here in the solution to highlight the formulaic structure of an if-then proof.

Group exercise #3: Solution

Proposition. Let x be an integer. Prove that $0|x$ if and only if $x = 0$.

Proof. We decompose the *if-and-only-if* statement into two *if-then* statements.

(a) We show that if $0|x$, then $x = 0$.

Let x be an integer such that $0|x$.

Then by definition of *divisible*, there is an integer a such that $0 \cdot a = x$.

But $0 \cdot a = 0$.

Hence $x = 0$.

(b) We show that if $x = 0$, then $0|x$.

Let $x = 0$.

Let a be any integer. (For example, take $a = 7$.) Then $x = a \cdot 0$.

Hence, there is an integer a such that $x = 0 \cdot a$.

Hence, $0|x$.

Remark 1. The color scheme corresponds to that of the previous two group exercises.

Remark 2. An *if-and-only-if* proof consists of two *if-then* proofs. Each uses the same *if-then* template (and same color-scheme) as in Group Exercises #1 and #2. Note that some green rows were skipped (as there was no definition to unravel for $x = 0$).

Group exercise #4: Solution

Proposition. An integer is odd if and only if it is the sum of two consecutive integers.

Proof. We decompose the *if-and-only-if* statement into two *if-then* statements.

(a) We show that if x is the sum of two consecutive integers, then x is an odd integer.

Let x be the sum of two consecutive integers.

So there is an integer a such that $x = a + (a + 1)$.

So $x = 2a + 1$

Hence, there is an integer a such that $x = 2a + 1$.

Hence, x is an odd integer.

(b) We show that if x is an odd integer, then x is the sum of two consecutive integers.

Let x be an odd integer.

Then by definition of *odd*, there is an integer a such that $x = 2a + 1$.

So we have $x = 2a + 1 = a + (a + 1)$.

Hence x is the sum of two consecutive integers.

Remark. An *if-and-only-if* proof consists of two *if-then* proofs. Each uses the same *if-then* template (and same color-scheme) as in Group Exercises #1 and #2.